# Aurora Stays Q&A

Retrieval-Augmented Generation (RAG) POC

14
October
2026

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# Executive Summary

- •Built a source-grounded RAG assistant for Aurora Stays hotel-policy questions.
- •Four policy PDFs are converted to structured Markdown, chunked, embedded, and indexed in FAISS.
- •Selected baseline: k=5, MMR lambda=0.50, fetch_k=20, CrossEncoder reranking.
- •Held-out result: 9/10 cases passed; composite score 0.941.
- •Average metrics: Faithfulness 0.993 | Recall 0.977 | Precision 0.808 | Relevancy 0.939.
- •Main remaining issue: the Accessible King benchmark question is underspecified; a focused clarification experiment raised precision to 0.961.

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# Business Problem Overview and Solution Approach

## Problem

- •Staff handle about 1,200 guest queries per week; roughly 40% require multiple policy documents.
- •Manual lookup takes 8–12 minutes and creates a 15–20% risk of incomplete or incorrect answers.
- •Property-specific fees, room features, cancellation rules, and accessibility guidance are easy to mix up.

## Solution approach

•Extract policy PDFs with Docling; preserve headings and tables as Markdown.
•Use token-aware chunking, domain embeddings, FAISS retrieval, MMR, and CrossEncoder reranking.
•Generate concise answers with GPT-4o-mini using a strict retrieved-context-only prompt.

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# Data Understanding

- •Knowledge base: four selectable-text PDF policy guides.
- •Coverage: room/property information; booking and cancellation; refunds and additional charges; conduct and accessibility.
- •Source structure includes headings, policy sections, Markdown tables, and a six-property matrix.
- •Properties represented: ASH-ATL, ASH-BOS, ASH-CHI, ASH-DEN, ASH-MIA, and ASH-SEA.
- •OCR is disabled because the supplied PDFs contain selectable text.
- •Chunks are enriched with source, section, property-code, and topic metadata for focused retrieval.

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# RAG - Chunking - Embedding Model

- •Embedding model: HarishMaths/Hotel-Policy-Embedding.
- •Tokenizer maximum length: 128 tokens.
- •Embeddings run on CPU with normalized vectors and progress rendering disabled.
- •The same model is used when building and loading the FAISS index.
- •Domain-specific embeddings improve matching for hotel-policy terms such as late checkout, pet fees, and room features.

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# RAG - Chunking - Chunker

- •Chunker: RecursiveCharacterTextSplitter.from_huggingface_tokenizer.
- •Chunk size: 512 tokens; overlap: 64 tokens.
- •Split priority: paragraph, line, heading, sentence, word, then character boundaries.
- •Atomic property rows are added for property-matrix entries so a single property and fee can be retrieved precisely.
- •Metadata is added after splitting to preserve section, topic, and property identity.

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# RAG - Chunking - Chunk Documents

- Total chunks created: 43, including atomic property rows.
- **Example focused chunk**

```
## Property-specific matrix | Property: ASH-MIA

ASH-MIA Brickell | Parking: Valet, $55 per night | Pets: $100 per stay; 2 pets max
```

- One-property rows reduce contamination from unrelated properties and broad policy sections.
- Each chunk carries source, section, property_codes, and topics metadata.

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# RAG - Vector Database

- Vector database: FAISS with 43 normalized document vectors.
- Index is persisted for the backend under backend/faiss_index.
- Final retrieval setting: k=5 chunks.
- Candidate search uses MMR lambda=0.50 and fetch_k=20.
- CrossEncoder ms-marco-MiniLM-L-6-v2 reranks candidates; topic and property metadata provide additional boosts.

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# RAG - Vector Database - Retrieval Function

Query: What are the parking fees at the Atlanta and Miami hotels?

- •Property routing identifies ASH-ATL and ASH-MIA; topic routing identifies parking.
  •Candidate chunks are retrieved with MMR, then reranked for direct relevance.
  **Relevant evidence**

  ASH-ATL Midtown: self parking, $28 per night.

  ASH-MIA Brickell: valet, $55 per night.

  •The final context contains the two requested property rows instead of unrelated policy sections.

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# RAG - Generation System - Load the LLM

- •Generator: OpenAI-compatible ChatOpenAI wrapper.
- •Model: gpt-4o-mini.
- •temperature=0.0 | top_p=1.0 | max_tokens=512.
- •API key and base URL are loaded from environment/configuration rather than hard-coded.
- •Deterministic settings favor repeatable, concise policy answers for evaluation and operations.

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# RAG - Generation System - Generation Prompt

Prompt inputs: user query + retrieved policy information.

- •Use only information explicitly supported by retrieved policy text.
- •Include relevant conditions, exclusions, limits, exceptions, and coverage details.
- •Focus on the question; omit unrelated content and do not invent facts.
- •If evidence is insufficient, return: The available policy documents do not contain enough information to answer this question.
- •Additional guards focus Advance Purchase, Accessible King, and late-checkout questions on their relevant evidence.

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# RAG - Generation System - RAG Function

- Pipeline: query -> retrieve top-k chunks -> generate grounded answer -> return answer and evidence.

Example query: What is the cancellation policy for a flexible reservation?

Example answer: Flexible rates can normally be canceled by 6:00 PM local property time one day before arrival without a room charge. Later cancellation or no-show may be charged one night plus tax; the confirmation controls if it differs.

•The response endpoint can return the answer alone or the answer plus relevant chunks for verification.

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# Evaluation - Load the Evaluation Data

- •Development evaluation file: data/evaluation.csv.
- •10 records with question, expected answer, and ground-truth relevant context.
  Sample question: What time can I check in at ASH-CHI Loop, and when do I need to be out?

  Expected answer: Standard check-in is 3:00 PM and standard check-out is 12:00 PM local property time; early arrival and late departure depend on availability and approval.

  •Multi-context records use <> separators so relevant policy sections can be evaluated together.

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# Evaluation - Define the Evaluation Model

- •Judge model: gpt-4o through the DeepEval OpenAIModel wrapper.
- •The generator remains gpt-4o-mini with temperature=0.0, top_p=1.0, and max_tokens=512.
- •The stronger judge model scores factual support, retrieval coverage, retrieval focus, and answer relevance.
- •Evaluations run synchronously so each test case has a stable, auditable metric result.

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# Evaluation - Define the Evaluation Metrics

Faithfulness >= 0.85: every important answer claim is supported by retrieved context.

- Contextual Recall >= 0.80: retrieved context contains information needed for the expected answer.

Contextual Precision >= 0.75: relevant context is prioritized over unrelated content.

Answer Relevancy >= 0.75: answer directly addresses the user question.

Composite = 0.35F + 0.25Recall + 0.20Precision + 0.20Relevancy.

A case passes only when all four metric thresholds are met.

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# Evaluation - Build and Run the Evaluation

- •Retrieval tuning matrix selected chunk_size=512, overlap=64, MMR lambda=0.50, and fetch_k=20.
- •Dev proxy coverage: 0.836; held-out test proxy coverage: 0.850.
- •Full DeepEval is the acceptance test; proxy coverage is used only to narrow candidates.
- •The evaluation creates LLMTestCase objects with input, expected output, actual output, retrieval context, and reference context.
- •Cases are classified as AUTO ANSWER, ASSISTED REVIEW, or MANUAL LOOKUP from the composite score.
- •Result: baseline k=5 is the selected configuration by highest composite score.

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# Evaluation - Load the Test Data

- •Held-out file: data/test.xlsx.
- •10 unseen questions covering cancellation, discounts, room occupancy, accessibility, parking, breakfast, rate changes, checkout, pets, and incidental holds.
  Sample question: How many people can stay in a Double Queen room?

  Expected answer: Up to 4 guests, subject to fire-code and property occupancy limits; recheck the room and rate if the group size changes.

  •The held-out set is used only for final comparison and selection of the baseline.

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# Evaluation - Build and Run the Evaluation

**Held-out DeepEval comparison (10 cases)**

- Baseline k=5 F=.993 R=.977 P=.808 AR=.939 Composite=.941 Pass=9/10
- Hybrid k=5 F=.995 R=.979 P=.787 AR=.922 Composite=.935 Pass=8/10
- Hybrid k=6 F=.993 R=.982 P=.781 AR=.929 Composite=.935 Pass=8/10

Selected: baseline k=5 because it has the highest weighted composite score.

Only remaining failure: Accessible King room inclusions; Contextual Precision=.451.

The benchmark question is underspecified. A focused clarification plus single-chunk retrieval scored Precision=.961 and passed all four metrics.

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